

ENG-654 · LECTURE 04

Inverse Kinematics of Wrist-Partitioned 6R Robots

Solve position with the arm, orientation with the wrist, and enumerate up to eight pose solutions

Running example: `custom_6R_new.urdf`. Transfer example: the PUMA 560.

Why a spherical wrist simplifies inverse kinematics

For a regular reachable target, the common intersection of axes 4–6 is the partition point O_w and organizes a finite solution tree.

 ≤ 4

arm solutions
place O_w

 $\times \leq 2$

wrist flips
match R_d

 ≤ 8

pose solutions
match T_d

$${}^0T_6(\mathbf{q}) = \underbrace{{}^0T_3(q_1, q_2, q_3)}_{\text{arm}} \underbrace{{}^3T_6(q_4, q_5, q_6)}_{\text{spherical wrist}}$$

The position solve reuses Lecture 03. Eight is a regular-case upper count: unreachable targets have none, tangent branches merge, wrist singularities create a continuous angle ambiguity, and limits or collisions remove otherwise mathematical solutions.

Six joint axes form an arm and a spherical wrist

Drag all six sliders. The red point is the spherical-wrist center.

1	2	3	4	5	6
<i>z</i>	<i>y</i>	<i>z</i>	<i>x</i>	<i>z</i>	<i>x</i>

Operational frame: use link_6. The separate tool0 branch is attached to link 3 and therefore is not the endpoint of this six-joint pose task.

☒ Labels

θ1

0°

θ2

0°

θ3

0°

θ5

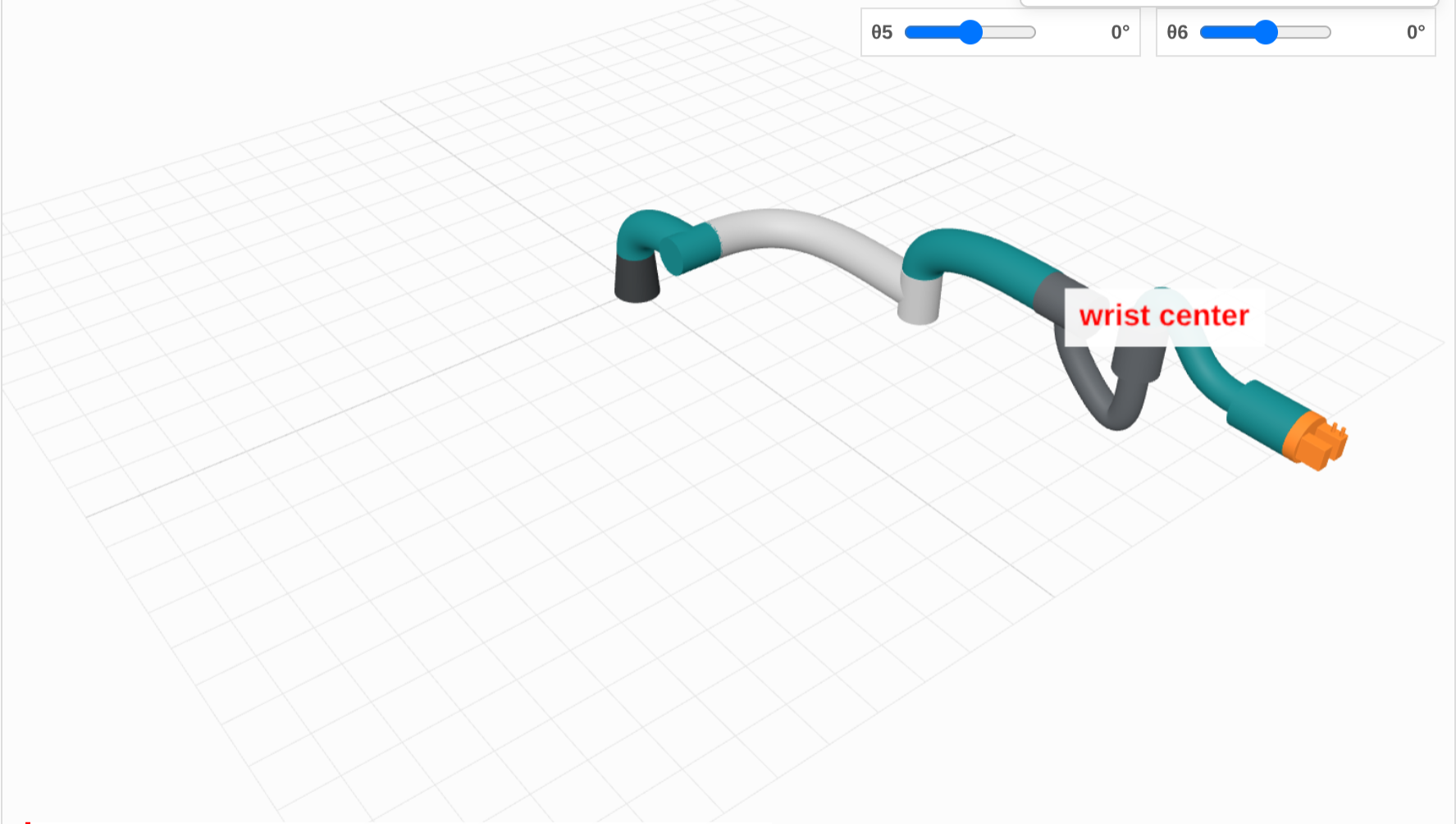
0°

θ6

0°

STL opacity

100%



Drag all six joints. The red wrist center moves only with $\theta_1-\theta_3$; $\theta_4-\theta_6$ change orientation about that point.

Concurrency belongs to axis lines—not to URDF origins

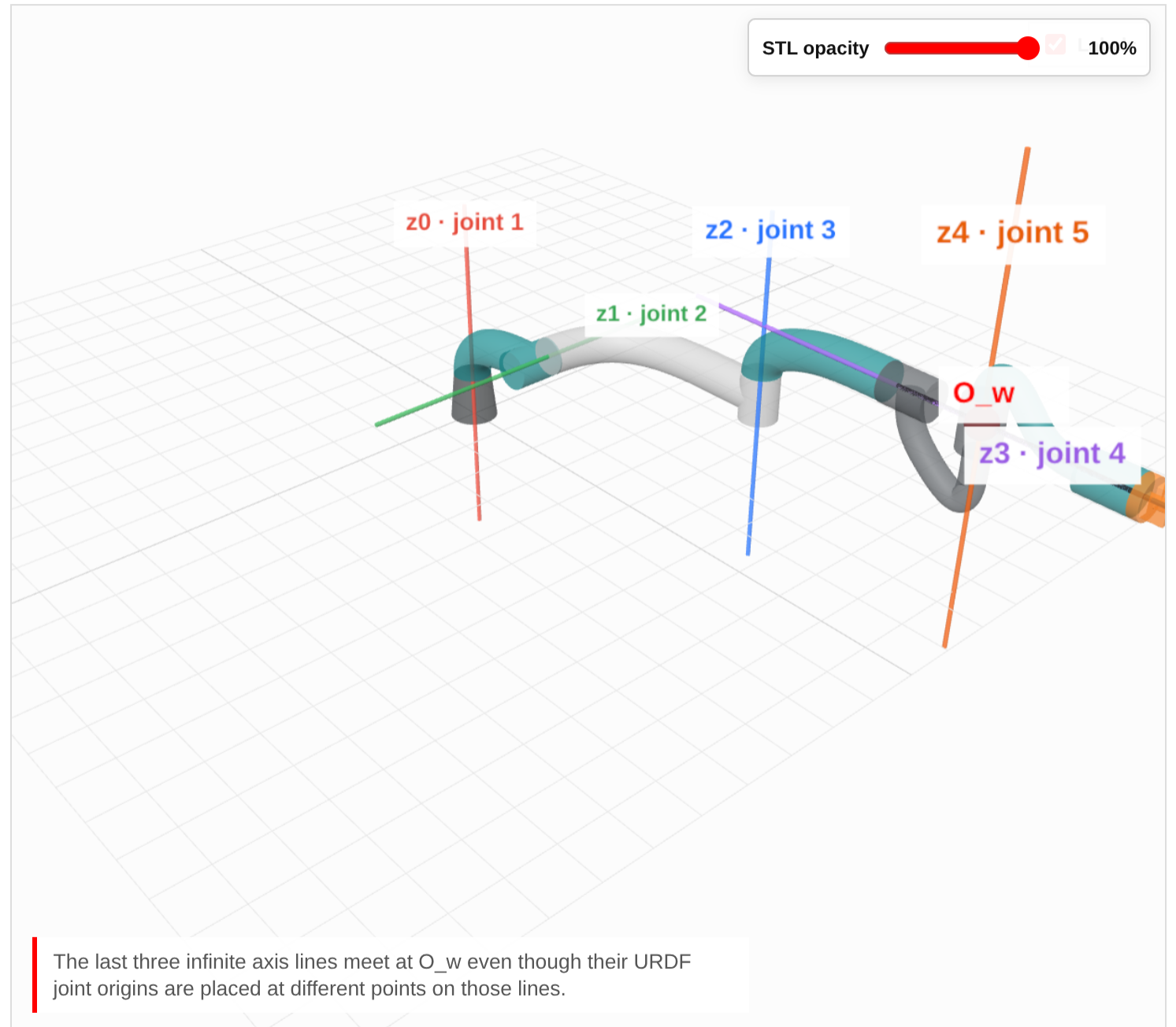
A revolute axis is an infinite line, not merely the point stored in the URDF joint origin.

axis 4 x -line through (4.5, 1.25, 1.75)

axis 5 z -line through (5.5, 1.25, 1.75)

axis 6 same x -line, stored at (7, 1.25, 1.75)

$$O_w = (5.5, 1.25, 1.75) \text{ m at home}$$



Rotating joints 4–6 changes orientation but leaves O_w fixed.

Construct each D–H frame from neighboring joint axes

Use standard D–H: $A_i = R_z(\vartheta_i)T_z(d_i)T_x(a_i)R_x(\alpha_i)$.

- 1** Axes 1–2: $a_1 = 1$, $d_1 = 1$, $\alpha_1 = -90^\circ$.
- 2** Axes 2–3: $a_2 = 2$, $d_2 = 1.25$, $\alpha_2 = +90^\circ$.
- 3** Axes 3–4 intersect after $d_3 = .75$, hence $a_3 = 0$.
- 4** Along axis 4, the 3–4 intersection to O_w is $d_4 = 1.5 + 1.0 = 2.5$.
- 5** Axes 4–5–6 meet at O_w , so $a_4 = a_5 = 0$ and $d_5 = 0$.
- 6** The link-6 origin lies $d_6 = 1.5$ along axis 6 from O_w .

The D–H table captures lengths, twists, and angle conventions

i	a_i [m]	α_i	d_i [m]	$\vartheta_i = q_i + \delta_i$
1	1.00	-90°	1.00	q_1
2	2.00	$+90^\circ$	1.25	q_2
3	0	$+90^\circ$	0.75	$q_3 + 90^\circ$
4	0	$+90^\circ$	2.50	$q_4 + 180^\circ$
5	0	$+90^\circ$	0	$q_5 + 180^\circ$
6	0	$+90^\circ$	1.50	$q_6 + 180^\circ$

Lengths come from axis-line geometry; offsets come from the selected D–H x_i directions at URDF home.

The same home pose has URDF and D–H descriptions

The robot stays in the same physical configuration; the joint-angle zeros and the attached frame axes differ.

URDF description

Joint coordinates are q_i , measured from the URDF home configuration. The endpoint frame is `link_6`.

$$\mathbf{q}_{\text{home}} = (0, 0, 0, 0, 0, 0)$$

D–H description

Angles are $\vartheta_i = q_i + \delta_i$, measured between the chosen D–H x -axes. At the same home configuration:

$$\boldsymbol{\vartheta}_{\text{home}} = \boldsymbol{\delta} = (0, 0, 90^\circ, 180^\circ, 180^\circ, 180^\circ)$$

$${}^0T_{6,U} = \left(\prod_{i=1}^6 A_i(q_i + \delta_i) \right) R_z(90^\circ)$$

The constant terminal rotation aligns the D–H end frame with the URDF `link_6` axes without moving its origin. With both corrections, the models describe the same pose for every $\mathbf{q}$.

Inverse kinematics must recover the entire pose

Every accepted IK branch must reproduce the complete homogeneous transform, not only its translation; the pose itself has six independent constraints.

$$T_d = \begin{bmatrix} .782996 & .589593 & .198234 & 5.112668 \\ .384659 & -.208506 & -.899201 & .585998 \\ -.488830 & .780324 & -.390052 & .715009 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} R_d & p_d \\ 0 & 1 \end{bmatrix}.$$

R_d	desired tool orientation
p_d	desired link_6 origin
in theory	$\det(R_d) = 1, R_d^\top R_d = I$ exactly.
in practice	Allow numerical tolerance: $ \det(R_d) - 1 \leq \varepsilon_{\det}$ and $\ R_d^\top R_d - I\ _F \leq \varepsilon_R$. Choose tolerances for the data precision; the displayed entries are rounded.

Orientation tells us where the wrist center must be

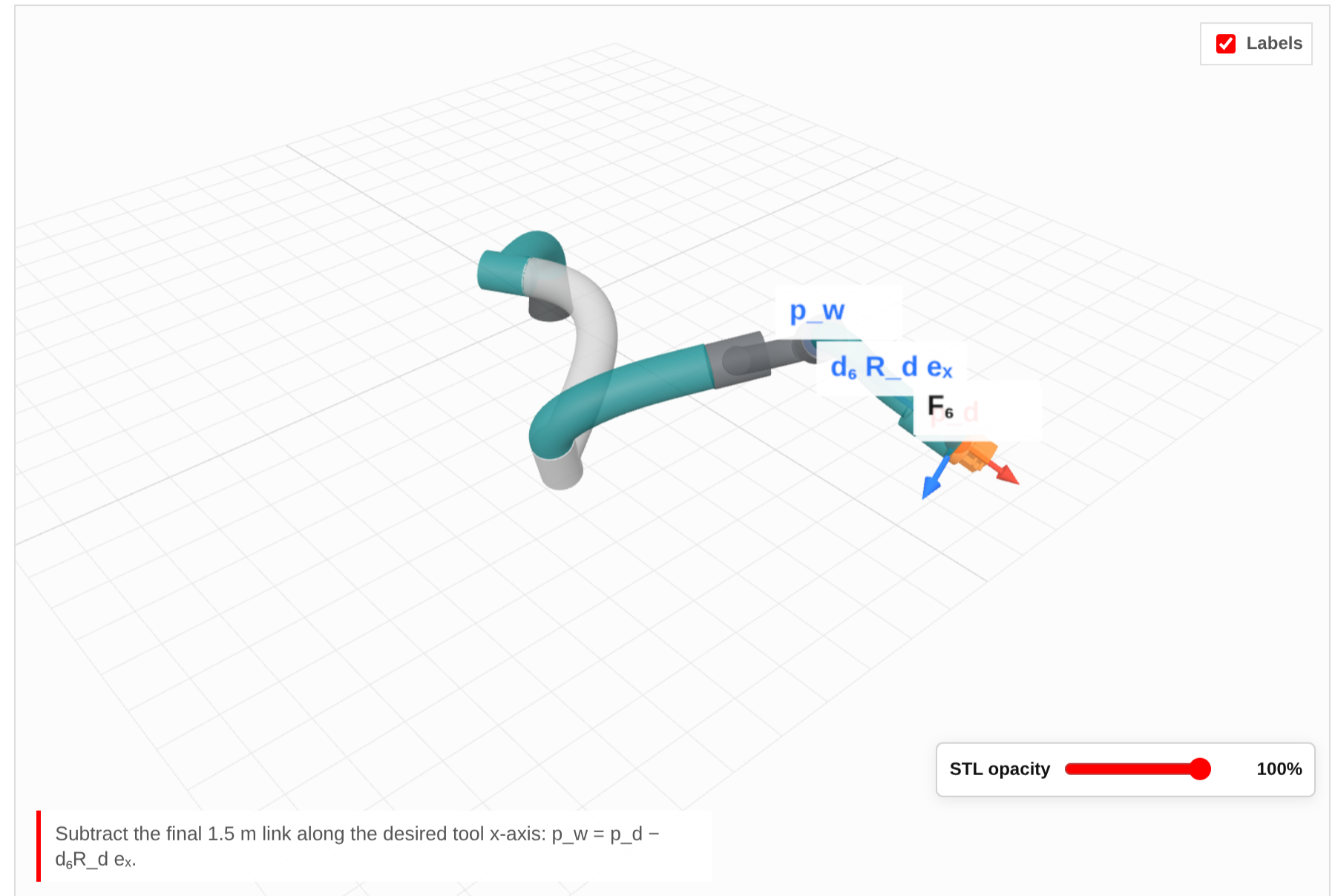
$$p_w = p_d - d_6 R_d e_x, \quad e_x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad d_6 = 1.5.$$

$$p_w = \begin{bmatrix} 5.112668 \\ .585998 \\ .715009 \end{bmatrix} - 1.5 \begin{bmatrix} .782996 \\ .384659 \\ -.488830 \end{bmatrix} = \begin{bmatrix} 3.938174 \\ .009009 \\ 1.448254 \end{bmatrix} \text{ m}.$$

Because $R_d e_x$ is the joint-6 axis direction, the wrist variables disappear from the position equation.

Walk backward from the tool frame to the wrist center

- 1 Read p_d and R_d from T_d .
- 2 Take the first column $R_d e_x$.
- 3 Move backward 1.5 m to obtain p_w .



The first three joints now solve a familiar position problem

Replace the Lecture-03 tool point by O_w ; the first three axes are unchanged.

01
Target

$$p_w = \begin{bmatrix} 3.938174 \\ .009009 \\ 1.448254 \end{bmatrix} \text{ m}$$

 $\rightarrow$
02
3R position IK

Apply Lecture-03 quartic elimination and back-substitution.

 $\rightarrow$
03
Branches

Retain every real (q_1, q_2, q_3) .

$$\begin{bmatrix} \mathbf{p}_w \\ 1 \end{bmatrix} = g_{\text{arm}}(q_1, q_2, q_3) \begin{bmatrix} \mathbf{p}_{w,0} \\ 1 \end{bmatrix}, \quad \mathbf{p}_{w,0} = \begin{bmatrix} 5.5 \\ 1.25 \\ 1.75 \end{bmatrix} \text{ m}$$

Here g_{arm} is the spatial product-of-exponentials displacement from the home configuration. It is distinct from the standard-D–H frame pose 0T_3 .

The wrist center admits four arm configurations

0R_3 is the rotation of frame $\{3\}$ relative to $\{0\}$. Here $\{3\}$ uses the URDF `link_3_d3` axes.

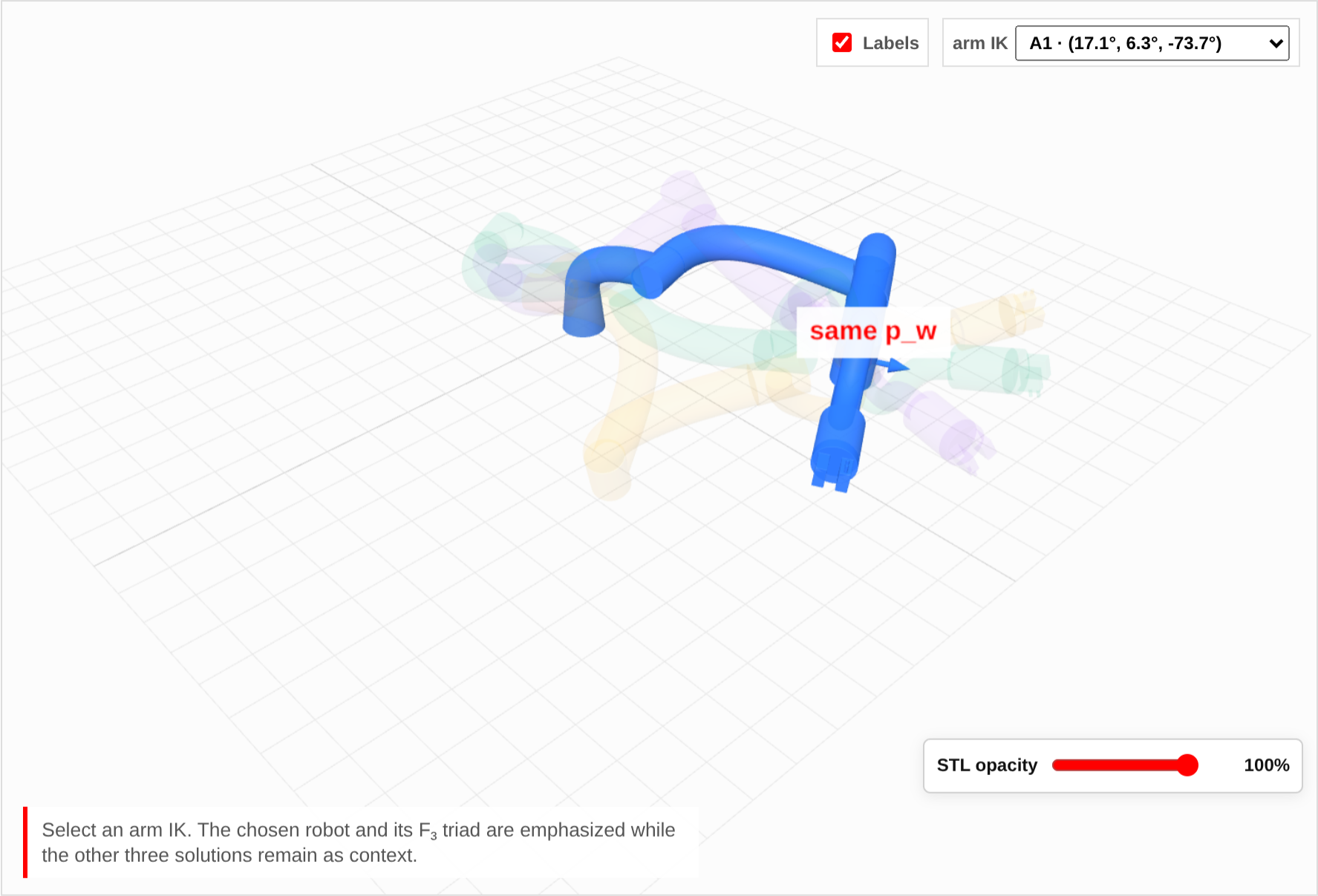
arm	q_1	q_2	q_3	z_3 orientation
A1	17.108°	6.318°	-73.729°	(.105, .032, .994)
A2	-151.329°	-164.628°	14.632°	(.233, .127, -.964)
A3	-107.813°	-149.154°	87.017°	(.157, .488, -.859)
A4	-60.000°	20.000°	120.000°	(.171, -.296, .940)

The wrist point is identical, but ${}^0R_3 = R_z(q_1)R_y(q_2)R_z(q_3)$ differs; every arm row therefore needs its own wrist solve.

Position agreement does not imply orientation agreement

position	p_w fixes every triad origin
orientation	0R_3 fixes the remaining wrist rotation

$${}^3R_6 = ({}^0R_3)^\top R_d$$



Each arm branch leaves a different rotation for the wrist

$$M_k = {}^3R_{6,k} = ({}^0R_{3,k})^\top R_d$$

- 1 Evaluate ${}^0R_{3,k}$ using arm row k .
- 2 Premultiply R_d by $({}^0R_{3,k})^\top$.
- 3 Decompose M_k using the wrist's $x - z - x$ sequence.

The spherical wrist reduces to an $x-z-x$ rotation

Write each elementary rotation with $c_i = \cos q_i$ and $s_i = \sin q_i$, then multiply in the order shown.

$$R_x(q_4) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_4 & -s_4 \\ 0 & s_4 & c_4 \end{bmatrix}$$

$$R_z(q_5) = \begin{bmatrix} c_5 & -s_5 & 0 \\ s_5 & c_5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_x(q_6) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_6 & -s_6 \\ 0 & s_6 & c_6 \end{bmatrix}$$

$$\begin{aligned} M = {}^3R_6 &= R_x(q_4) R_z(q_5) R_x(q_6) \\ &= \begin{bmatrix} c_5 & -s_5c_6 & s_5s_6 \\ s_5c_4 & -s_4s_6 + c_4c_5c_6 & -s_4c_6 - c_4c_5s_6 \\ s_4s_5 & s_4c_5c_6 + c_4s_6 & -s_4c_5s_6 + c_4c_6 \end{bmatrix}. \end{aligned}$$

middle angle	$m_{11} = c_5$
first angle	$(m_{21}, m_{31}) = s_5(c_4, s_4)$
last angle	$(-m_{12}, m_{13}) = s_5(c_6, s_6)$

A regular spherical wrist has two equivalent flips

$$\begin{aligned}s_5 &= \pm \sqrt{m_{12}^2 + m_{13}^2}, \\ q_5 &= \operatorname{atan} 2(s_5, m_{11}), \\ q_4 &= \operatorname{atan} 2(m_{31}/s_5, m_{21}/s_5), \\ q_6 &= \operatorname{atan} 2(m_{13}/s_5, -m_{12}/s_5).\end{aligned}$$

$$(q_4, q_5, q_6) \leftrightarrow (q_4 + \pi, -q_5, q_6 + \pi)$$

Singularity: if $s_5 = 0$, the two discrete flips collapse into a continuum. For $q_5 = 0$, only $q_4 + q_6$ is identifiable; for $q_5 = \pi$, only $q_4 - q_6$ is identifiable.

Either wrist flip completes the same desired pose

A4 · flip 1

$(-145^\circ, 50^\circ, -110^\circ)$

A4 · flip 2

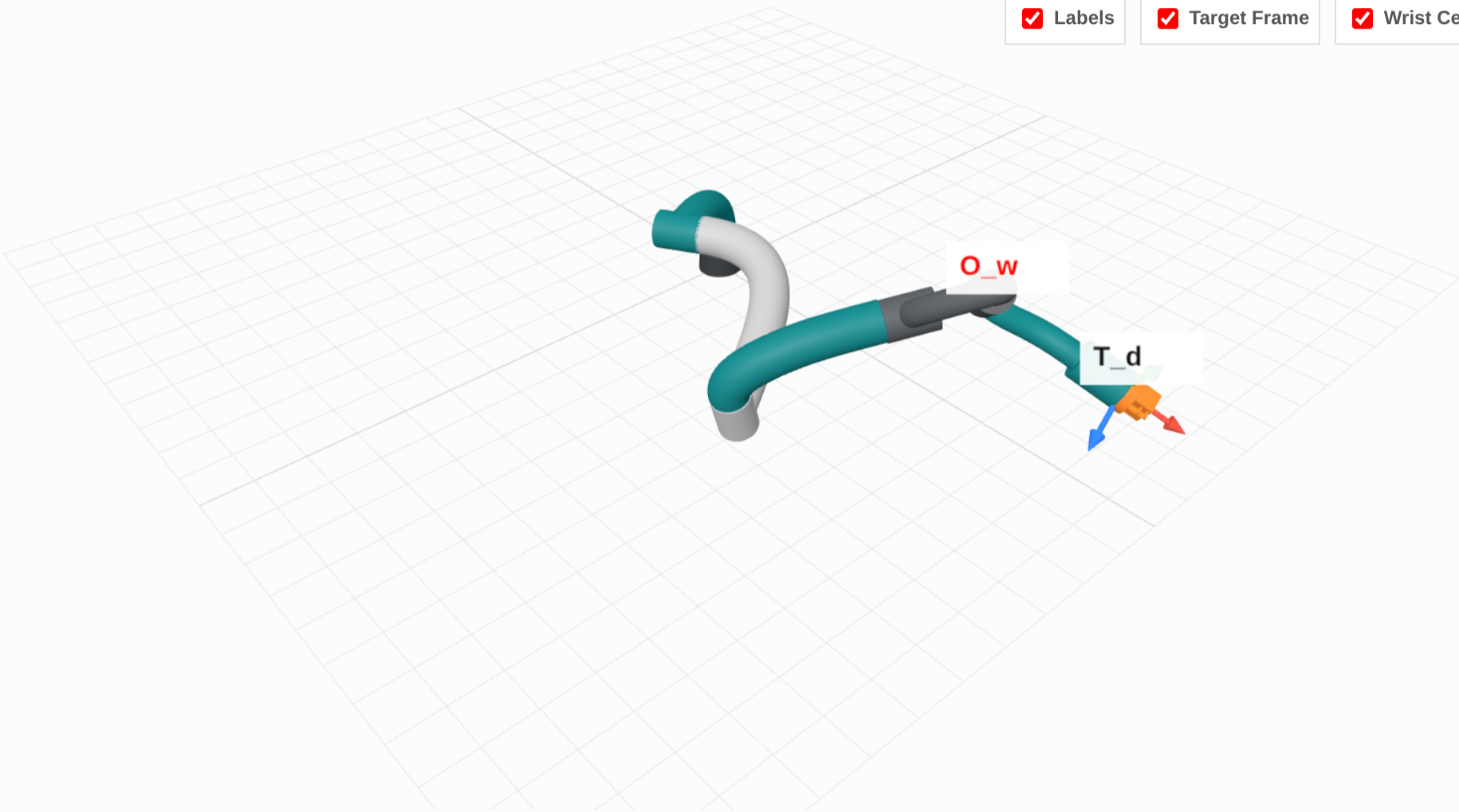
$(35^\circ, -50^\circ, 70^\circ)$

☒ Labels

☒ Target Frame

☒ Wrist Center

solution IK 7 · arm 4 · flip 1 ▾



Choose any arm branch and either wrist flip. Both flips preserve the complete desired pose.

STL opacity 100%

This regular target produces eight pose solutions

solution	q_1 [°]	q_2 [°]	q_3 [°]	q_4 [°]	q_5 [°]	q_6 [°]
1	17.108	6.318	-73.729	-23.207	82.925	118.503
2	17.108	6.318	-73.729	156.793	-82.925	-61.497
3	-151.329	-164.628	14.632	-78.522	-45.784	9.391
4	-151.329	-164.628	14.632	101.478	45.784	-170.609
5	-107.813	-149.154	87.017	-72.063	-50.137	-32.928
6	-107.813	-149.154	87.017	107.937	50.137	147.072
7	-60.000	20.000	120.000	-145.000	50.000	-110.000
8	-60.000	20.000	120.000	35.000	-50.000	70.000

Forward kinematics of every row agrees with T_d to better than 2×10^{-7} in every matrix entry.

Eight joint configurations realize this one end-effector pose

IK 1–2 arm A1, two wrist flips

IK 3–4 arm A2, two wrist flips

IK 5–6 arm A3, two wrist flips

IK 7–8 arm A4, two wrist flips

☒ Labels

IK 1°

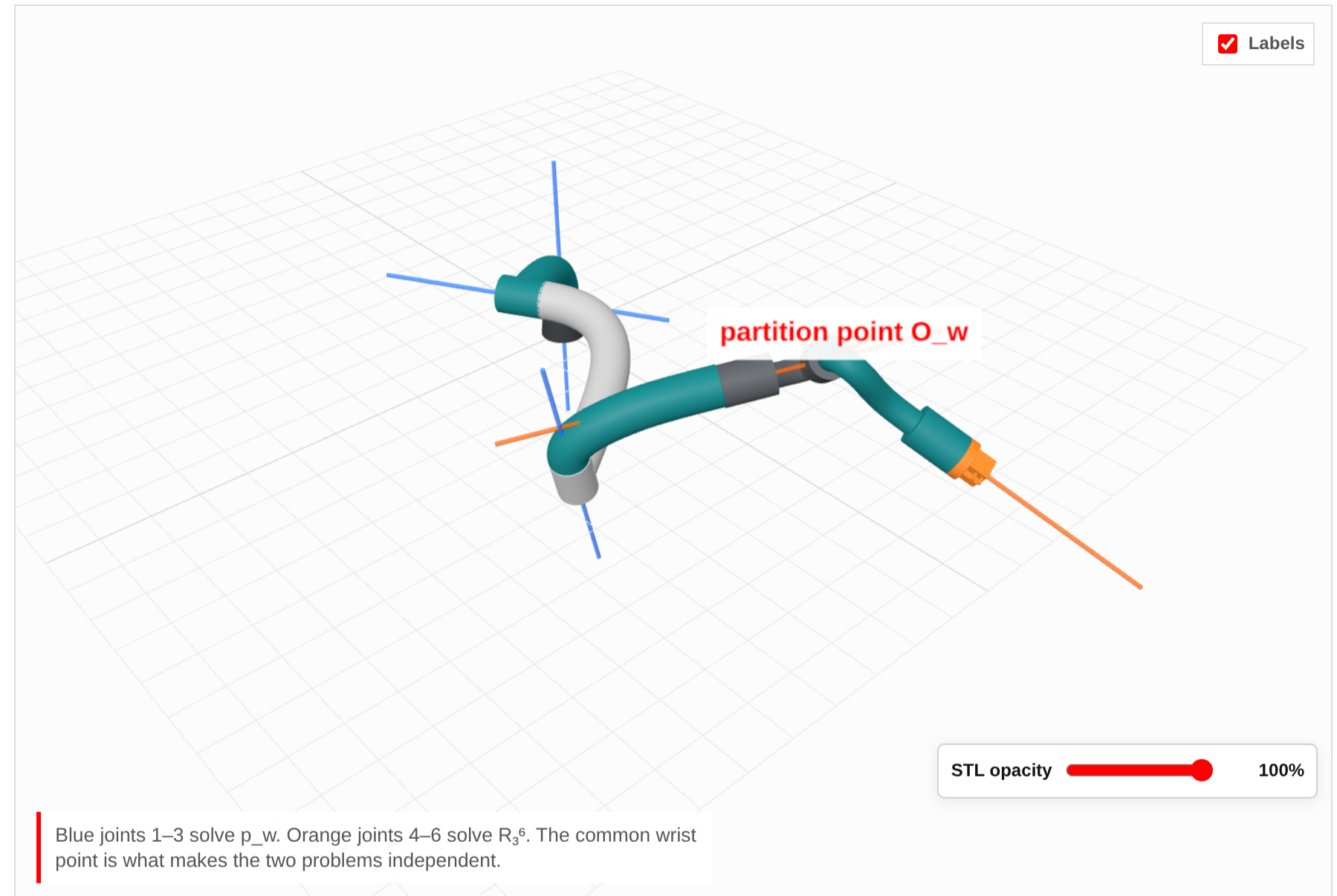
same T_d

STL opacity 100%

Drag the IK selector through all eight exact solutions. Every configuration reproduces the same position and orientation.

Wrist partition organizes the solution into five steps

- 1 Compute $p_w = p_d - d_6 R_d e_x$.
- 2 Solve all 3R arm branches for p_w .
- 3 For each branch, compute $M = ({}^0R_3)^\top R_d$.
- 4 Extract both $x - z - x$ wrist flips.
- 5 Enforce limits and verify ${}^0T_6 = T_d$.



The PUMA 560 exposes the same position–orientation partition

Explore the course STL model and track the common wrist center.

arm	joints 1–3 position O_w
wrist	axes 4–6 intersect at O_w
tool	link7, the child of joint 6

☒ Labels

q1

0°

q2

0°

q3

0°

q4

0°

q5

0°

q6

0°

Move the PUMA joints. Joints 4–6 rotate about one common wrist point; only q₁–q₃ move that point.

STL opacity

100%

Axis-line distances provide the constants for the arm solve

base height 0.6718 m

axes 2–3 $\sqrt{.4318^2 + .0203^2} = .43228$ m

shoulder offset 0.1501 m along the parallel axes

forearm 0.4331 m to the wrist center

$$p_w = p_d - 0.0558R_de_z$$

The final link lies along the tool z -axis. Subtracting it removes joints 4–6 from the position equation.

$$p_w = f_{\text{arm}}(q_1, q_2, q_3)$$

The numerical constants differ from the custom robot, but the two-stage reasoning is identical.

Forward kinematics creates the pose that inverse kinematics must recover

$$q_d = (30^\circ, -35^\circ, 45^\circ, 40^\circ, -50^\circ, 60^\circ)$$

$$p_w = \begin{bmatrix} .436419 \\ .078646 \\ -.019019 \end{bmatrix} \text{m}$$

$$T_d = \begin{bmatrix} .112612 & -.525432 & .843351 & .483478 \\ -.939575 & -.332458 & -.081670 & .074089 \\ .323291 & -.783194 & -.531121 & -.048656 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

This target is nonsingular and produces four mathematical arm branches before joint limits are applied.

Shoulder and elbow choices generate four ways to reach p_w

Apply the Lecture-03 Paden–Kahan construction to the first three PUMA axes.

arm	q_1	q_2	q_3	geometric branch
P1	30.000°	−35.000°	45.000°	right shoulder · elbow 1
P2	30.000°	−77.351°	129.617°	right shoulder · elbow 2
P3	170.431°	−139.617°	129.617°	left shoulder · elbow 2
P4	170.431°	−97.266°	45.000°	left shoulder · elbow 1

$${}^3R_{6,k} = ({}^0R_{3,k})^\top R_d, \quad k = 1, \dots, 4$$

Each PUMA arm branch admits two spherical-wrist flips

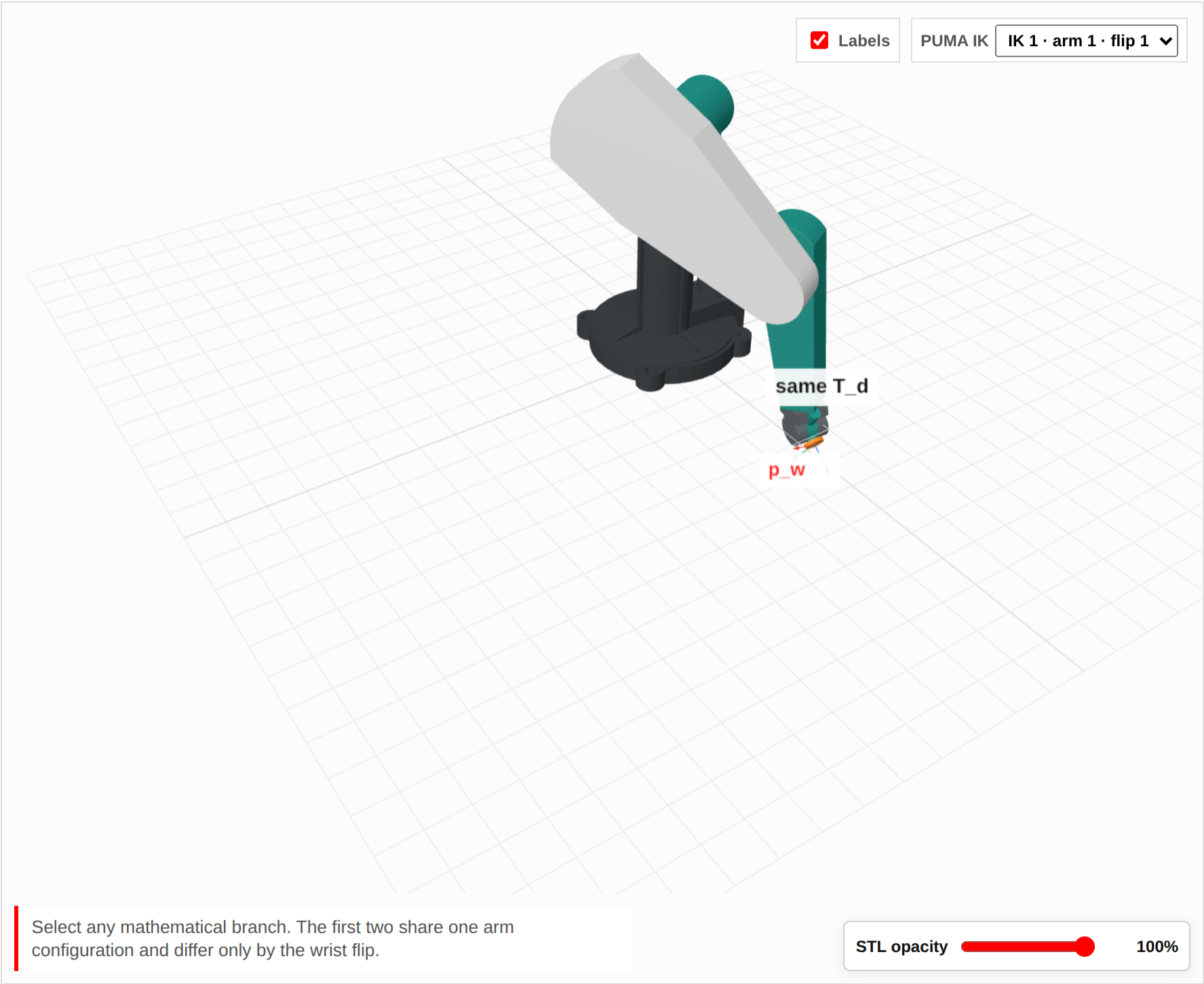
The final axes form a z – y – z -type wrist under the URDF sign conventions.

solution	q_1 [°]	q_2 [°]	q_3 [°]	q_4 [°]	q_5 [°]	q_6 [°]
1	30.000	–35.000	45.000	–140.000	50.000	–120.000
2	30.000	–35.000	45.000	40.000	–50.000	60.000
3	30.000	–77.351	129.617	–90.226	29.499	178.601
4	30.000	–77.351	129.617	89.774	–29.499	–1.399
5	170.431	–139.617	129.617	–4.609	47.947	–111.487
6	170.431	–139.617	129.617	175.391	–47.947	68.513
7	170.431	–97.266	45.000	–31.539	6.549	–83.205
8	170.431	–97.266	45.000	148.461	–6.549	96.795

These are all mathematical branches modulo 360° . The URDF joint limits retain only solution 2 for this target.

All eight mathematical branches reproduce the same pose

position	the red p_w is unchanged
orientation	the colored target triad is unchanged
limits	filter only after enumeration



A 7R arm has one internal degree of freedom at a fixed pose

7

joint variables

− 6

pose constraints

= 1

redundancy parameter

$$\text{rank } J(\mathbf{q}^*) = 6 \implies \dim_{\mathbf{q}^*} \mathcal{S}(T_d) = 7 - 6 = 1$$

Locally around a regular solution, the redundant IK set is a one-dimensional self-motion curve. At rank loss, its local dimension or branch structure can change.

Fixing q_3 defines a parameterized 6R IK subproblem

$$q_3 = \lambda$$

At a chosen value of λ , the remaining variables are

$$\bar{q} = (q_1, q_2, q_4, q_5, q_6, q_7) \in \mathbb{R}^6.$$

$$F_\lambda(\bar{q}) = {}^0T_7(q_1, q_2, \lambda, q_4, q_5, q_6, q_7) = T_d.$$

Six unknown joint angles now satisfy the six independent constraints of a full pose.

Solving this parameterized 6R problem analytically produces branches $\bar{q}_k(\lambda)$ of the 7R inverse kinematics.

A joint variable becomes a geometric parameter of the reduced robot

Original iiwa · 7R

Seven revolute coordinates; every twist is fixed.

i	θ_i	d_i	a_i	α_i
1	q_1	d_1	a_1	α_1^0
2	q_2	d_2	a_2	α_2^0
3	q_3	d_3	a_3	α_3^0
4	q_4	d_4	a_4	α_4^0
5	q_5	d_5	a_5	α_5^0
6	q_6	d_6	a_6	α_6^0
7	q_7	d_7	a_7	α_7^0

$q_3 = \lambda$
→
 $\bar{\alpha}_2 = \lambda$

Parameterized member · 6R

Six revolute coordinates; geometry is indexed by λ .

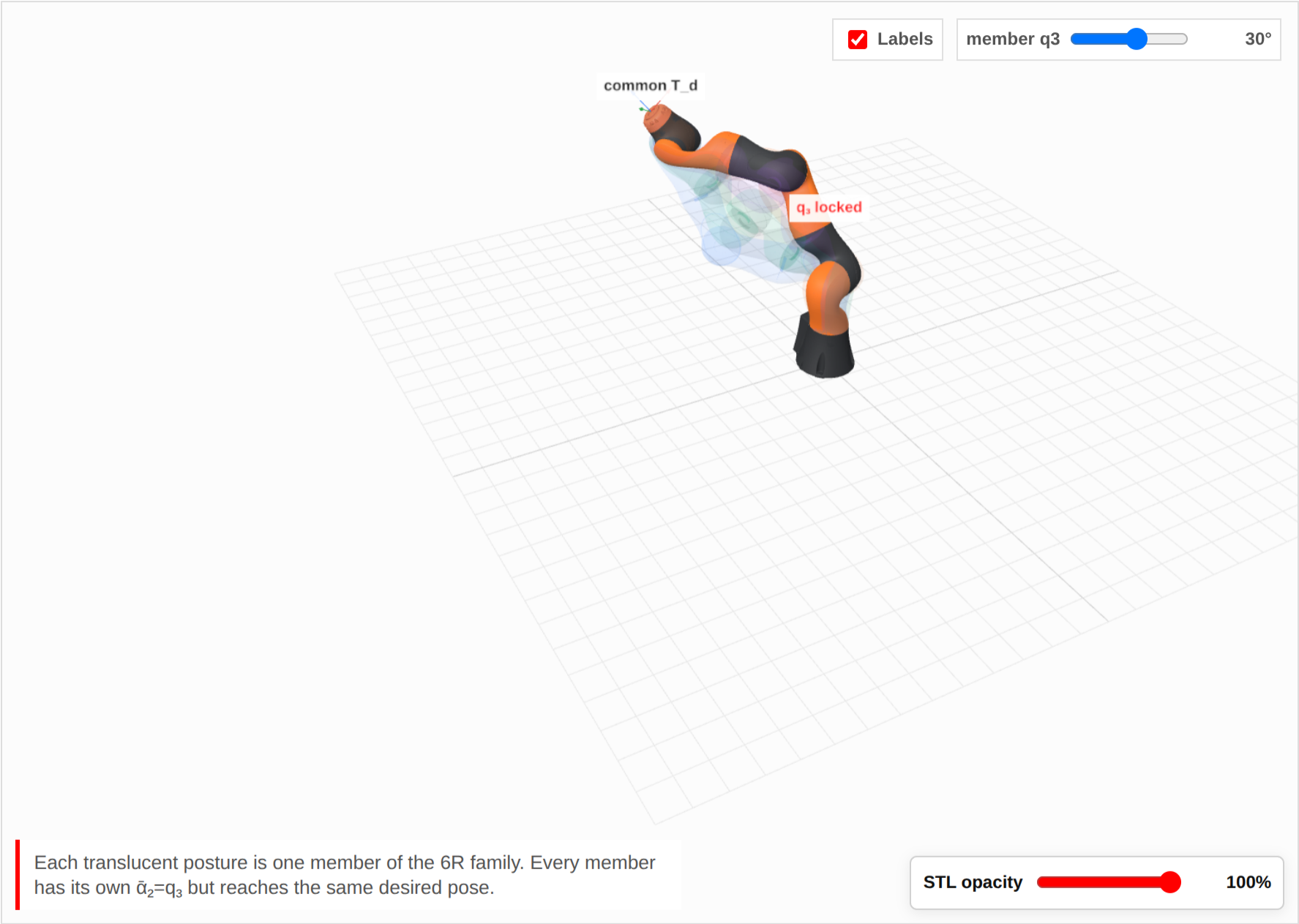
j	iiwa axis	$\bar{\theta}_j$	$\bar{d}_j$	$\bar{a}_j$	$\bar{\alpha}_j$
1	1	q_1	$\bar{d}_1$	$\bar{a}_1$	$\bar{\alpha}_1$
2	2	q_2	$\bar{d}_2$	$\bar{a}_2$	$\bar{\alpha}_2 = q_3 = \lambda$
3	4	q_4	$\bar{d}_3$	$\bar{a}_3$	$\bar{\alpha}_3$
4	5	q_5	$\bar{d}_4$	$\bar{a}_4$	$\bar{\alpha}_4$
5	6	q_6	$\bar{d}_5$	$\bar{a}_5$	$\bar{\alpha}_5$
6	7	q_7	$\bar{d}_6$	$\bar{a}_6$	$\bar{\alpha}_6$

With the displayed reduced-frame convention, locking joint 3 removes q_3 from the coordinate vector and places its chosen value in the geometry: $\bar{\alpha}_2 = \lambda$. A different frame convention may add a fixed angular offset.

The 7R IK set is assembled from 6R solution sets indexed by q_3

$$\mathcal{S}_{7R}(T_d) = \bigcup_{\lambda \in I} \mathcal{S}_{6R}(T_d; q_3 = \lambda)$$

choose	$q_3 = \lambda$
freeze	joint 3 becomes geometry
solve	the remaining six joints
vary	λ to recover the 7R family

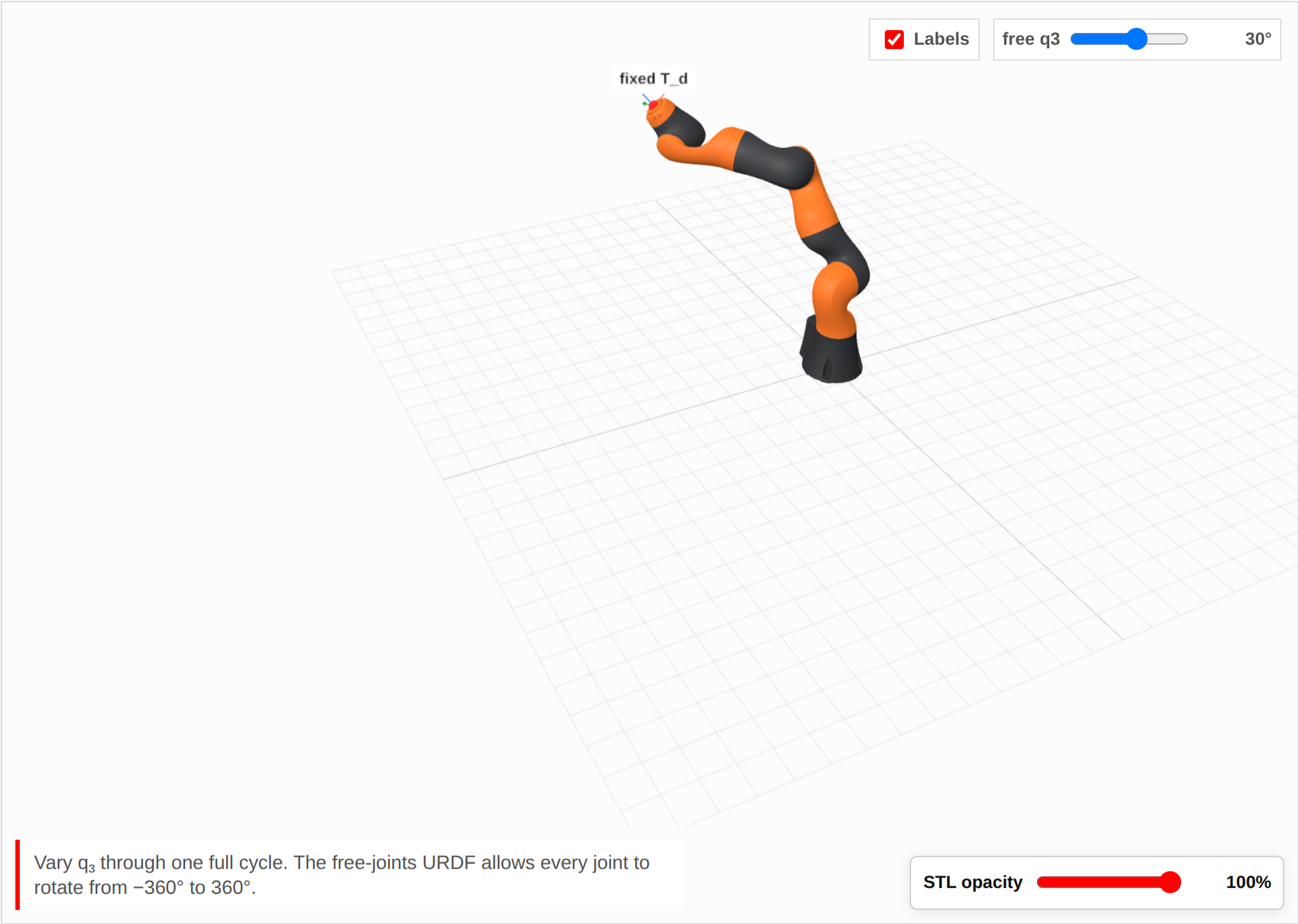


The translucent postures sample parameter values along a feasible solution branch. Their reduced D–H tables differ in $\tilde{\alpha}_2$, yet all reproduce the same T_d .

Changing q_3 moves the elbow while the tool pose stays fixed

$$q_d = (20, -35, 30, 50, -40, 45, 60)^\circ$$

free parameter	$q_3 \in [-180^\circ, 180^\circ]$ in the displayed cycle
dependent joints	$q_1, q_2, q_4, q_5, q_6, q_7$
invariant	the complete T_d , not only p_d



The visualization follows one numerically continued, forward-kinematics-verified branch; it is not an exhaustive enumeration of every connected component. The model gives each revolute joint a $[-360^\circ, 360^\circ]$ range.

The redundancy parameter is free only inside feasible intervals

- 1 Choose $q_3 = \lambda$ within its mechanical range.
- 2 Solve the resulting parameterized 6R IK subproblem and retain every real branch.
- 3 Continue each branch as λ varies; this traces a self-motion curve.
- 4 Stop or change branch at workspace boundaries, singularities, collisions, or joint limits.

$$\mathcal{S}(T_d) = \bigcup_{\lambda \in I} \{(q_1, q_2, \lambda, q_4, q_5, q_6, q_7)_k\}$$

Robot geometry determines the right inverse-kinematics decomposition

- 1 A spherical wrist separates a 6R pose problem into arm position and wrist orientation.
- 2 The custom robot and PUMA use the same logic with different axis geometry and tool directions.
- 3 Four arm branches and two wrist flips can create eight mathematical 6R solutions.
- 4 For a 7R iiwa, fixing one redundancy parameter exposes a family of parameterized 6R solves.
- 5 Joint limits, singularities, and collisions determine which mathematical branches are usable.

Bridge to Lecture 05: branch counts and local continuation change where the Jacobian loses rank; singularities explain where the inverse ceases to behave regularly.

Reference: J. J. Craig, *Introduction to Robotics: Mechanics and Control*, manipulator kinematics and inverse kinematics chapters.